The Kamrad–Ritchken trinomial tree and its equivalence to FTCS

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1 Setting and motivation

A CRR binomial tree for American options is already available (`quantlib.american`; see the note *American Options by Longstaff-Schwartz*), together with four PDE-based pricers (FTCS, BTCS, CN, and CN-PSOR for American). What does the trinomial tree add?

Not pricing capability: the binomial CRR already prices Americans; CN-PSOR already prices them more accurately. The contribution of this note is conceptual — it makes *explicit the structural equivalence between lattice methods and finite-difference methods*. The earlier notes allude to this equivalence ("FTCS and the trinomial are the same calculation written in two languages"); this note formalises it.

Two additional benefits:

- The finite-difference/lattice benchmark (`benchmark_phase3.py`) compares all the project's pricers on the American put. The trinomial gives one more independent point of comparison.
- The trinomial of Boyle [2], generalised by Kamrad and Ritchken [1], has a stretching parameter λ that does not exist in the binomial. This adds a tuning knob to the family of lattice methods; the benchmark uses the default $\lambda = 3$ only.

2 Tree structure

We work in log-coordinates $x = \ln(S/S_0)$, so the GBM $dS = rS\,dt + \sigma S\,dW$ becomes

$$dX = \nu\,dt + \sigma\,dW, \qquad \nu = r - \tfrac{1}{2}\sigma^2.$$

The trinomial tree discretises this process with time step $\Delta t = T/N$ and space step Δx (to be chosen). At each node, the next state branches three ways:

$$x \longrightarrow \begin{cases} x + \Delta x & \text{with probability } p_u \\ x & \text{with probability } p_m \\ x - \Delta x & \text{with probability } p_d \end{cases}$$

with $p_u + p_m + p_d = 1$. The key structural features:

- **Recombining**: an up-then-down move arrives at the same state as a stay-then-stay move. After n time steps, only $2n + 1$ distinct states are reachable (not 3^n). This is what makes the backward induction tractable: $O(N^2)$ total work, not $O(3^N)$.
- **Symmetric in space**: the same Δx for up and down. This simplifies the moment-matching algebra and makes the tree agree with the centred-difference FTCS scheme, as we will see.

The Kamrad–Ritchken step size. We take

$$\Delta x = \sigma\sqrt{\lambda\,\Delta t}, \qquad \lambda \geq 1. \tag{1}$$

The parameter λ is the *stretching parameter*. Kamrad and Ritchken write the step as $\lambda_{\mathrm{KR}}\,\sigma\sqrt{\Delta t}$, with $p_m = 1 - 1/\lambda_{\mathrm{KR}}^2$; the λ of this note is the square of theirs, $\lambda = \lambda_{\mathrm{KR}}^2$. Three values are commonly used:

λ	Tree type	p_m
1	Binomial-equivalent ($p_m = 0$)	0
2	Conservative trinomial	1/2
3	Gauss–Hermite ($\lambda_{\mathrm{KR}} = \sqrt{3}$; default)	2/3

We will see in §5 that λ is structurally the inverse of the FTCS Fourier number: $\lambda = 1/(2\alpha)$. The constraint $\lambda \geq 1$ is exactly the FTCS CFL condition $\alpha \leq 1/2$ written as a tree constraint.

3 Probabilities by moment matching

The probabilities (p_u, p_m, p_d) are determined by matching the first two moments of the discrete one-step distribution to those of the continuous GBM:

$$\mathbb{E}[\Delta X] \;=\; p_u\,\Delta x - p_d\,\Delta x \;=\; \nu\,\Delta t, \tag{2}$$

$$\mathbb{E}[(\Delta X)^2] \;=\; (p_u + p_d)\,\Delta x^2 \;=\; \sigma^2\,\Delta t + O(\Delta t^2). \tag{3}$$

Combined with the normalisation $p_u + p_m + p_d = 1$, this is a linear 3×3 system. Substituting (1) and solving:

$$\boxed{\;p_u = \frac{1}{2\lambda} + \frac{\nu\sqrt{\Delta t}}{2\sigma\sqrt{\lambda}}, \quad p_m = 1 - \frac{1}{\lambda}, \quad p_d = \frac{1}{2\lambda} - \frac{\nu\sqrt{\Delta t}}{2\sigma\sqrt{\lambda}}.\;} \tag{4}$$

Reading. The first term $1/(2\lambda)$ in p_u and p_d encodes the variance of the step. The second term $\pm\nu\sqrt{\Delta t}/(2\sigma\sqrt{\lambda})$ is the asymmetry that accommodates the drift. The middle probability p_m is determined purely by λ.

Positivity constraints.

- $p_m \geq 0 \iff \lambda \geq 1$.
- $p_d \geq 0$ (and $p_u \geq 0$ when $\nu < 0$) $\iff |\nu|\sqrt{\lambda\,\Delta t} \leq \sigma \iff \dfrac{\lambda\,\nu^2\,\Delta t}{\sigma^2} \leq 1$, i.e. $\lambda \leq \sigma^2/(\nu^2\Delta t)$: the grid Péclet condition $|\nu|\Delta x/\sigma^2 \leq 1$ of the FTCS note. For project parameters ($|\nu| \sim 0.03$, $\sigma = 0.20$, $\Delta t \leq 0.02$) $\nu^2\Delta t/\sigma^2 \leq 4.5 \times 10^{-4}$, so with $\lambda = 3$ the LHS is at most 1.35×10^{-3}, far below 1.
- $p_u, p_m, p_d \leq 1$ is then automatic.

For the project's parameters, all three probabilities lie comfortably in $(0, 1)$.

4 Backward induction

With probabilities fixed, the algorithm is standard.

Initialisation at maturity ($n = N$). The states at time T are $S_j = S_0\,e^{j\,\Delta x}$ for $j \in \{-N, -N+1, \ldots, N\}$. For each, set $V_j^N = \phi(S_j)$ where ϕ is the payoff function ($(K-S)^+$ for a put, $(S-K)^+$ for a call).

Backward step ($n = N-1, \ldots, 0$). For each $j \in \{-n, \ldots, n\}$,

$$V_j^n = e^{-r\Delta t}\big[\, p_u\, V_{j+1}^{n+1} + p_m\, V_j^{n+1} + p_d\, V_{j-1}^{n+1} \,\big] \tag{5}$$

for the European option, or

$$V_j^n = \max\Big\{\, \phi(S_j),\;\; e^{-r\Delta t}\big[\, p_u\, V_{j+1}^{n+1} + p_m\, V_j^{n+1} + p_d\, V_{j-1}^{n+1} \,\big] \,\Big\} \tag{6}$$

for the American option. The price is V_0^0.

Cost. At step n there are $2n+1$ active nodes. Total work is $\sum_{n=0}^{N-1}(2n+1) = N^2$ multiply-add operations, i.e. $O(N^2)$ time. With in-place overwrite of a single length-$(2N+1)$ array, memory is $O(N)$. Identical to CRR.

5 The equivalence with FTCS

This is the structural insight that motivates including the trinomial among the finite-difference methods. The European backward step (5) can be rewritten as

$$V_j^n = e^{-r\Delta t} p_u\, V_{j+1}^{n+1} + e^{-r\Delta t} p_m\, V_j^{n+1} + e^{-r\Delta t} p_d\, V_{j-1}^{n+1}. \tag{7}$$

Compare with the FTCS update for the transformed BS PDE $V_\tau = (\sigma^2/2)V_{xx} + \nu V_x - rV$ (see the FTCS note, with $\Delta\tau = \Delta t$):

$$V_j^{n+1} = (\alpha+\beta)V_{j+1}^n + (1 - 2\alpha - r\Delta t)V_j^n + (\alpha-\beta)V_{j-1}^n \tag{8}$$

where $\alpha = (\sigma^2/2)\Delta t/\Delta x^2$ and $\beta = \nu\Delta t/(2\Delta x)$.

Note. Equation (8) marches forward in τ (and hence backward in t since $\tau = T - t$), so the time index on the LHS is $n+1$ while the RHS uses n. Equation (7) marches in calendar time, with the LHS at n and RHS at $n+1$. Both schemes go from later to earlier in calendar time. The structural form is the same: a three-point linear combination of the values one time level away.

Coefficient identification. Substituting $\Delta x = \sigma\sqrt{\lambda\Delta t}$ into the FTCS coefficients:

$$\begin{aligned} \alpha &= \frac{\sigma^2\Delta t/2}{\lambda\,\sigma^2\,\Delta t} = \frac{1}{2\lambda}, \\ \beta &= \frac{\nu\,\Delta t}{2\,\sigma\sqrt{\lambda\,\Delta t}} = \frac{\nu\,\sqrt{\Delta t}}{2\,\sigma\,\sqrt{\lambda}}. \end{aligned}$$

Then the FTCS coefficients become

$$\begin{aligned} \alpha+\beta &= \frac{1}{2\lambda} + \frac{\nu\sqrt{\Delta t}}{2\sigma\sqrt{\lambda}} = p_u, \\ \alpha-\beta &= \frac{1}{2\lambda} - \frac{\nu\sqrt{\Delta t}}{2\sigma\sqrt{\lambda}} = p_d, \\ 1-2\alpha-r\Delta t &= 1 - \frac{1}{\lambda} - r\Delta t = p_m - r\Delta t. \end{aligned}$$

The *un-discounted* off-diagonal coefficients agree exactly: $p_u = \alpha+\beta$ and $p_d = \alpha-\beta$ as identities. The two schemes differ in *how the discount factor is distributed* across the recurrence:

- Trinomial multiplies the entire bracketed expression by $e^{-r\Delta t}$, applying the discount to all three terms uniformly.
- FTCS embeds the discount only in the diagonal coefficient ($1 - 2\alpha - r\Delta t = p_m - r\Delta t$), leaving the off-diagonals undiscounted.

The middle coefficient comparison: $e^{-r\Delta t} p_m - (p_m - r\Delta t) = r\Delta t/\lambda + O(\Delta t^2)$, which is $O(\Delta t)$. The off-diagonal comparison $e^{-r\Delta t} p_u - (\alpha + \beta) = -r\Delta t\, p_u + O(\Delta t^2)$ is also $O(\Delta t)$. The two schemes therefore agree at the leading order $O(1)$ and differ at $O(\Delta t)$, which is the same order as the local truncation error of both schemes. They are equivalent in the sense that they share the same order of convergence and the same limiting solution, but their numerical outputs at finite Δt differ by a small ($O(\Delta t)$) amount.

Conclusion.

$$\boxed{\text{Trinomial KR with stretching } \lambda \;\sim\; \text{FTCS with } \alpha = \tfrac{1}{2\lambda}} \tag{9}$$

in the asymptotic sense: both schemes have $O(\Delta t)$ local truncation error, both converge to the same BS solution at the same rate, and their numerical outputs differ by $O(\Delta t)$ (the same order as the error itself). The two methods are not literally identical, but they are structurally the same scheme written in two languages.

The CFL constraint, in tree language. Equation (9) also explains why $\lambda \geq 1$ is a hard constraint: $\lambda = 1$ corresponds to $\alpha = 1/2$, the FTCS stability boundary. Going below ($\lambda < 1 \iff \alpha > 1/2$) violates the CFL condition. In tree language this manifests as $p_m < 0$: a probability that is no longer a probability. The same phenomenon, two different symptoms.

Why $\lambda = 3$. With $\lambda = 3$ the probabilities are $(p_u, p_m, p_d) = (1/6, 2/3, 1/6) + O(\sqrt{\Delta t})$ and the nodes are 0 and $\pm\sigma\sqrt{3\Delta t}$. These are exactly the nodes and weights of the three-point Gauss–Hermite rule for $\mathcal{N}(0, \sigma^2\Delta t)$, which integrates polynomials up to degree 5 exactly. (The weights coincide with those of Simpson's rule, but the relevant fact is the Gauss–Hermite one: Simpson's rule is exact for a uniform weight, not a Gaussian one.) The moments of the step ΔX make this concrete. For any λ,

$$\mathbb{E}[\Delta X] = \nu\Delta t, \qquad \mathbb{E}[\Delta X^2] = \sigma^2\Delta t,$$

which match the Gaussian log-step to leading order, while

$$\mathbb{E}[\Delta X^3] = \lambda\sigma^2\nu\Delta t^2, \qquad \mathbb{E}[\Delta X^4] = \lambda\sigma^4\Delta t^2,$$

against $3\sigma^2\nu\Delta t^2 + O(\Delta t^3)$ and $3\sigma^4\Delta t^2 + O(\Delta t^3)$ for the Gaussian. Only $\lambda = 3$ matches the third and fourth moments. The global convergence rate nevertheless remains $O(\Delta t)$, because the non-smooth payoff (the kink at the strike) limits the achievable order regardless, and for this payoff the better moment match does not give a smaller error constant either. The λ sweep of `validate_trinomial.py` (European call, $S = K = 100$, $r = 0.05$, $\sigma = 0.2$, $T = 1$, $n = 1000$ steps) gives errors 2.06×10^{-3} ($\lambda = 1$), 6.4×10^{-4} ($\lambda = 1.5$), 1.09×10^{-3} ($\lambda = 2$), 1.99×10^{-3} ($\lambda = 3$) and 3.80×10^{-3} ($\lambda = 5$): $\lambda = 3$ is no better than $\lambda = 1$, and $\lambda = 1.5$ is the best of the five on this problem.

6 Convergence and comparison

The trinomial converges to the BS price for European options, and to the true American price for American options, at rate $O(1/N)$ in the number of steps. This is the same rate as CRR binomial. The kink-related limitation discussed in the Crank–Nicolson note (the payoff is not C^1) caps both lattice methods at first order regardless of the underlying scheme accuracy.

Method	Cost (American)	Convergence
CRR binomial	$O(N^2)$	$O(1/N)$
Trinomial KR	$O(N^2)$	$O(1/N)$
CN-PSOR	$O(N^{5/2}\log\epsilon^{-1})$, $M \propto N$	$O(\Delta x^2) + O(\Delta\tau)$

The binomial and trinomial have the same asymptotic order, and the trinomial's better moment match does not make its errors smaller at fixed N. For the American put, where both trees appear in the benchmark, the trinomial's error is about three times larger: 9.97×10^{-4} against 3.17×10^{-4} at $n = 1600$ (`convergence.csv`), and 1.70×10^{-3} against 6.14×10^{-4} at $n = 1000$ (`pricing_canonical.csv`), both measured against the lattice-consensus reference of `benchmark_phase3.py`. CN-PSOR is the most accurate of the three on these runs but the most expensive. In `convergence.csv`, where $M = \max(N/4, 50)$ (so $\Delta\tau$ is fixed for $N \leq 200$ and halves with Δx beyond), its error falls from 2.69×10^{-2} to 1.22×10^{-4} over $N = 50$–800 (ratios 3.9, 3.4, 3.6, 4.7), close to second order in N on that range; at fixed N its time error is first order in $\Delta\tau$ (see the PSOR note). The benchmark script `benchmark_phase3.py` compares them numerically on a common American put benchmark.

7 Implementation outline

A single module is enough. `quantlib.trinomial` (Python) and `quant::pde::trinomial` (C++) provide:

- `trinomial_european_call(S, K, r, sigma, T, *, n_steps, lambda_param=3.0)`
- `trinomial_european_put(S, K, r, sigma, T, *, n_steps, lambda_param=3.0)`
- `trinomial_american_put(S, K, r, sigma, T, *, n_steps, lambda_param=3.0)`

Why no `trinomial_american_call`: for a stock with no dividends, the American call price equals the European call price (Merton 1973), so the function would duplicate the European version.

The internal structure is a single helper function `_trinomial_backward` that performs the backward induction, parametrised by an option-type flag (`european_call`, `european_put`, `american_put`) that determines the terminal payoff and whether the early-exercise $\max\{\cdot, \phi\}$ check is applied at each step.

8 Validation strategy

Five tests:

1. **Cross-validation (European)** against Black–Scholes closed form on a parameter grid. Tolerance 5×10^{-3} at $N = 2000$; the largest observed error is 1.42×10^{-3}.
2. **Cross-validation (American)** against an embedded CRR binomial reference (the same approach as the CN-PSOR test). Tolerance $\sim 5 \times 10^{-3}$.
3. **First-order convergence in N**: doubling N should halve the error, distinguishing the trinomial from any hypothetical second-order scheme.
4. **Lambda sweep**: at fixed $N = 1000$, $\lambda \in \{1, 1.5, 2, 3, 5\}$ (including the degenerate binomial-equivalent $\lambda = 1$) all give prices within 5×10^{-3} of Black–Scholes (largest error 3.80×10^{-3}, at $\lambda = 5$), with a spread below 10^{-2}. This confirms that the choice of λ within the valid range $\lambda \geq 1$ is a tuning preference, not a correctness condition.

5. **Input validation**: $\lambda < 1$ raises (it would violate $p_m \geq 0$); $n_steps < 1$ raises; non-positive parameters raise.

The fourth test is the operational confirmation that the λ-parameter is a tuning knob within a valid range, not a correctness-determining parameter.

References

References

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